2D CAM FOR CNC · IN THE BROWSER

COPO

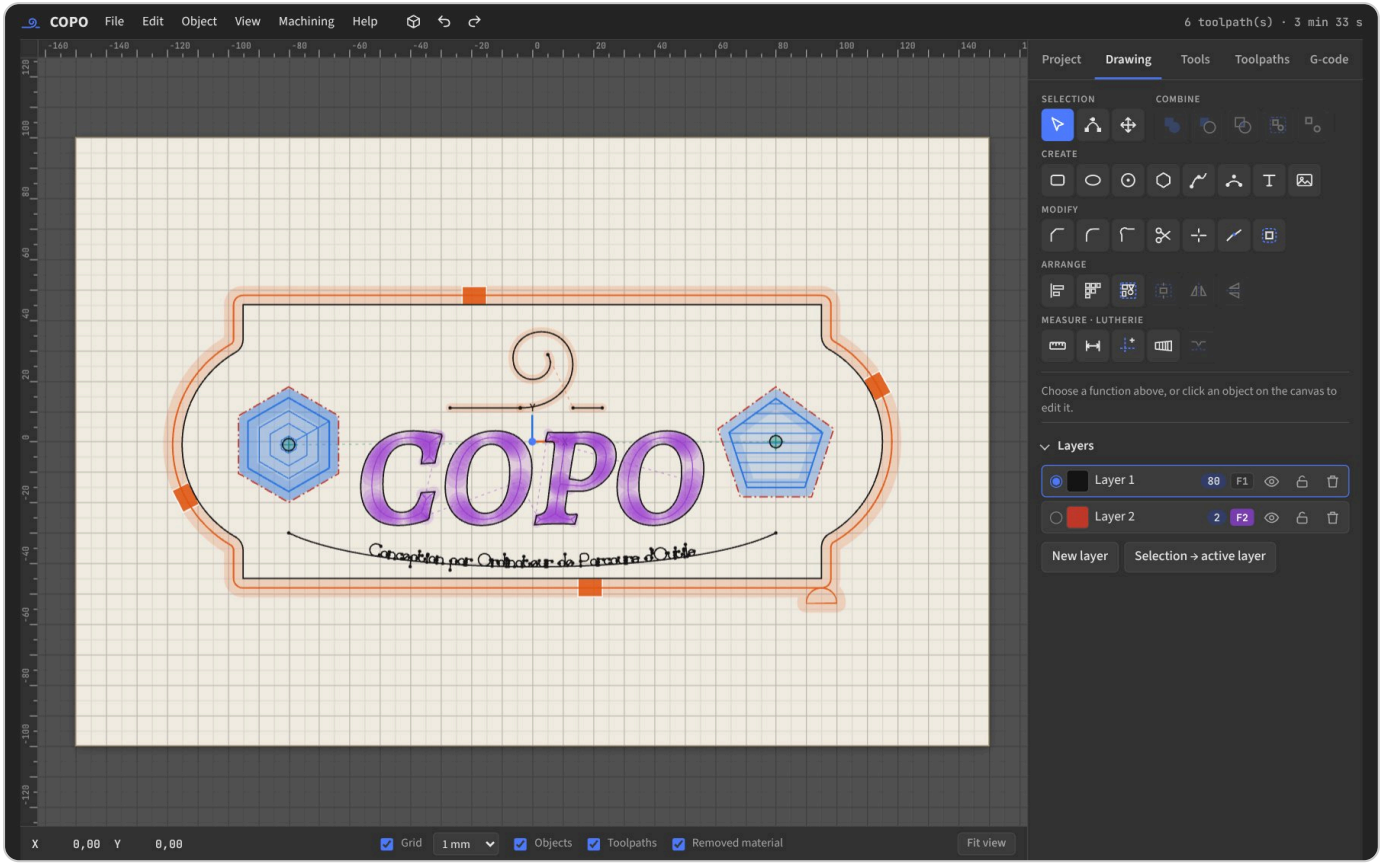
COPO draws your parts as vectors, calculates toolpaths (profile, pocket, V-carving, drilling) and writes the G-code for your machine. Everything runs in the browser, with no installation and no account.

`Version 1.0` `GPL v3` `Windows · macOS · Linux`

`Chrome · Edge · Firefox` `Français · English`

Open the app
`https://gbulleryahen-hash.github.io/copo/`

GitHub repository (code, download)
`https://github.com/gbulleryahen-hash/copo`



The sample project: COPO text V-carved, a line of single-line text on a curve, a decorative flourish with a ball nose bit, two drilled holes, a cut-out with tabs and two pockets on side 2. Toolpaths are shown in color on the stock.

At a glance

Vector drawing
Rectangle, ellipse, polygon, star, arc, Bézier pen, text, nodes, chamfers, fillets, booleans, layers, dimensions.

2D machining
Outside, inside or on-the-line profile, ring or raster pocket, rest machining, drilling.

V-carving
Along the medial axis, like Aspire: the bit rises up to the tip of every corner.

Tool entries
Back-and-forth ramp, continuous ramp, arc, ramped arc, helix. Allowance and finishing pass.

Tabs
Spaced automatically or placed by hand, flat or triangular.

Nesting
Places several parts on the sheet following their true shape, with rotations.

Single-line fonts
About sixty included (Hershey, EMS, Relief…) for fast one-pass engraving.

Double-sided
Layers per side, flip around X or Y, pause to turn the part over.

3D preview
Material removal simulation, side 1 and side 2, with animated playback.

G-code
GRBL, manual GRBL (G53), Carbide Motion, Onefinity, LinuxCNC, Mach3/4, UCCNC, MASSO or a custom post-processor. G2/G3 arcs.

Setup sheet
Printable workshop sheet: 2D and 3D views, origin, tools, toolpaths, speeds, checks.

Millimetres or inches
Drawing, tools and feeds in mm or inches (fractions available), G-code in G21 or G20.

French and English
Interface and user guide in both languages, chosen from the Help menu.

Free software
GNU GPL v3 licence: use, modify and share COPO freely.

Your first job in six steps

1. **Set up the project**
   Project tab: stock size and material, X0 Y0 origin point, Z zero (top of stock or table), units and machine.

2. **Draw or import**
   Draw with the icons of the Drawing tab, import an SVG or DXF (**File** › **Import an SVG or DXF…**) or type some text.

3. **Check your tools**
   Tools tab: diameter, spindle speed, feed rate, plunge rate, pass depth and stepover of each bit.

4. **Create the toolpaths**
   Select shapes on the canvas, open the **Toolpaths** tab, choose Profile, Pocket, V-carving or Drilling, adjust, then click **Calculate**.

5. **Check in 3D**
   View › **3D machining preview** shows the machined stock. Play the job back to spot a wrong depth or tool.

6. **Save the G-code**
   G-code tab: choose the post-processor, then **Save G-code**. Always do a dry run before cutting.

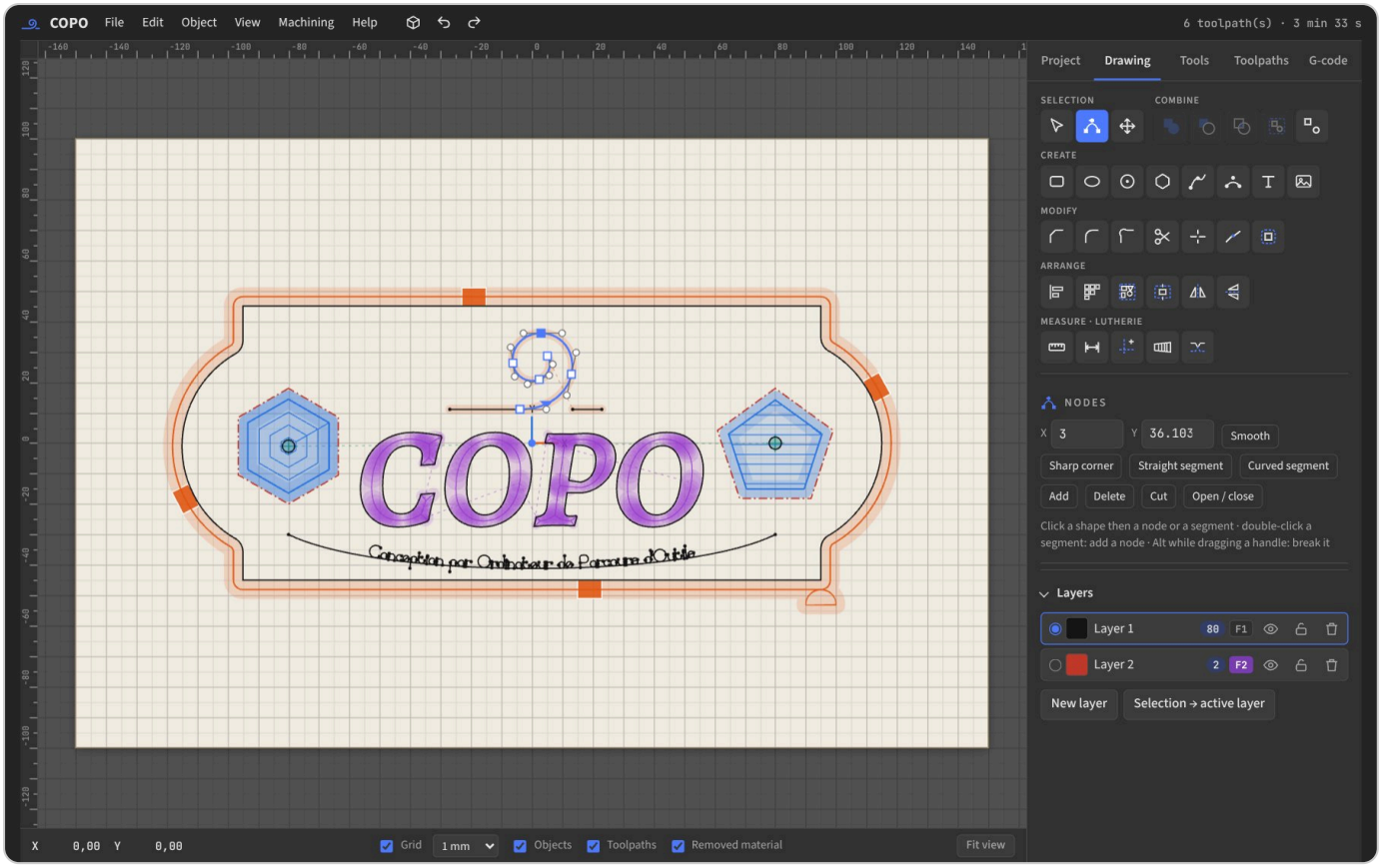
The interface

The window reads from left to right, in the order you work.

- **Menu bar**: File, Edit, Object, View, Machining and Help. The Help menu opens this guide, switches the language and shows the licence.
- **Drawing tab**: at the top, a grid of icons gathers every drawing function (Selection, Combine, Create, Modify, Arrange, Measure · Lutherie), each with a tooltip and a shortcut letter. Just below appears what you need: the settings of the chosen tool (with a hint and a small diagram), the form of the clicked function (Align, Array copy, Nesting, Images…), or the Selection block when an object is selected. Greyed icons need a selection first. Layers stay at the bottom. Esc closes the open function.
- **Workspace**: the stock in beige, rulers in millimetres, the grid and the origin (X and Y arrows).
- **Right panel** with five tabs: **Project**, **Drawing**, **Tools**, **Toolpaths** and **G-code**. In the Project, Tools and Toolpaths tabs, small diagrams show what each setting does (depth, allowance, cut direction, tool entry, tabs, bit shape, Z zero…) and update with your choices.
- **Status bar**, at the bottom: cursor coordinates relative to the origin, grid, snapping, toolpath and removed-material display.

A floating bar appears under the selection: **Duplicate**, **Group**, **Weld**, **Machine…** and **Delete**. The project is saved automatically in the browser; **File › Save project** writes a copy to a file.

Drawing



The Nodes tool on the decorative flourish: the options bar turns a node into a sharp or smooth corner, a segment into a line or a curve, and adds, deletes or cuts a node.

Shapes and paths

- **Rectangle** `R`, with sharp, rounded or chamfered corners. **Ellipse** `E` (Shift: circle) and **Circle from center** `C`.
- **Polygon and star** `O`: number of sides or points, inner radius.
- **Pen** `P`: click for a corner point, click-drag for a curve point, click the first point to close, double-click to finish. The tool settings take absolute, relative or polar coordinates: type the values then press `Enter`.
- **3-point arc** `A`: start, end, then the bulge.
- **Nodes** `N`: move nodes and handles, double-click a segment to add a node, `Alt` while dragging a handle to break it.

Editing

- **Chamfer** `K` and **Fillet** `F` on one corner, or on every sharp corner from the **Corners** block.
- **Trim** `X` deletes the part of a vector between two intersections; **Cut** `U` splits a vector where you click.

- **Weld, Subtract, Intersect** for boolean operations; **Join** and **Close** for open vectors; inside or outside **Offset contour**.
- **Align and distribute**: choose the reference in “Align to”. *The selection*: a single object aligns to the stock, several align to each other. *The first selected object*: it stays put and the others align to it; Distribute places them after it, in the order you selected them, with the given spacing. *The stock*: aligns to the stock edges and distributes across its full width or height.
- **Array copy** in a grid or a circle, **mirrors**, **rotation**, **stacking order**.
- **Exact position and size**: **Selection** block, with a reference point chosen among nine (corners, middles, center).

Snapping, guides and dimensions

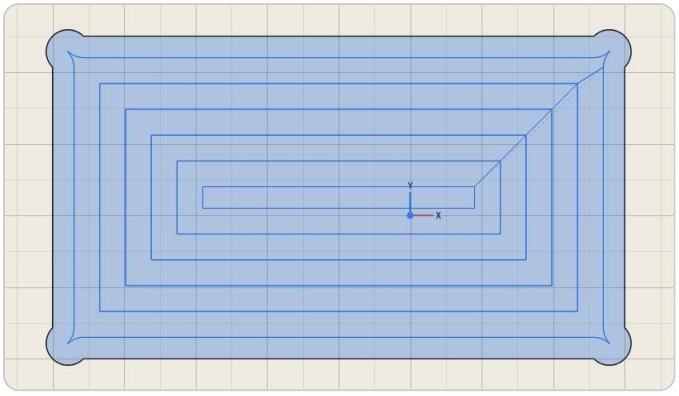
Object snapping finds endpoints, midpoints, intersections, centers and tangents; each one can be turned on or off in the **View** menu. Drag from a ruler to create a magnetic guide. The **Measure** tool `M` gives a distance and the **Dimension** tool `D` places a dimension that stays on the canvas.

Layers, images and files

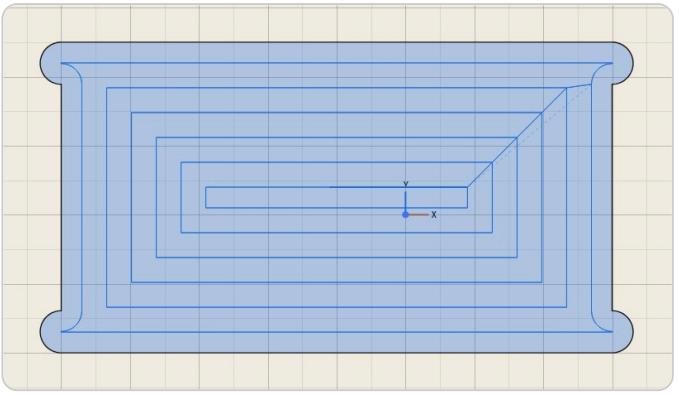
- **Layers**: stroke color, visibility, lock and machined side (F1 top, F2 bottom).
- **Reference images** (**File › Insert an image…** or drag and drop): move, resize and rotate them with the handles to trace over them. They are never machined.
- **Vectorize an image** (Drawing panel › Images, image selected): COPO traces the outlines of the image. Set the *threshold* (anything darker becomes the shape; “Auto threshold” works it out for you) and the *smoothing* (softer outlines, small details removed), tick *Invert* to outline light areas. The preview updates live; **Create outlines** adds closed, grouped shapes where the image is, ready for a pocket, a profile or V-carving. A sharp, high-contrast image (logo, line drawing, silhouette) gives the best result.
- **SVG**: import in millimetres with curves kept; export the whole drawing or the selection.
- **DXF** (text/ASCII, any version): lines, polylines with arcs, circles, arcs, ellipses, splines and blocks. The file's unit is respected (mm, cm, inches…); a file without a unit is read as millimetres. Touching segments are joined into outlines, closed when possible, and each DXF layer becomes a COPO layer. Text, dimensions, hatches and 3D objects are skipped (the message says which).

Corners for tenons and mortises: dogbone and T-bone

A round bit leaves a fillet of its radius in corners, so a square tenon will not fit its mortise. The relief is drawn on the shape, corner by corner, so it only goes where it is needed. Take the **Relief** tool `B` (Modify group), choose *Dogbone* (the bit moves diagonally to touch the sharp corner) or *T-bone* (the relief extends the longer side, less visible) and the **bit diameter**, then click near the corners to relieve. The same settings are in the **Corners** block of the Drawing panel. Corners are picked one by one, as for chamfers and fillets. Relief only works between two straight segments; its diameter is 0.1 mm larger than the bit so it fits without forcing. Then machine the shape as usual (pocket, inside or outside profile) with that bit.

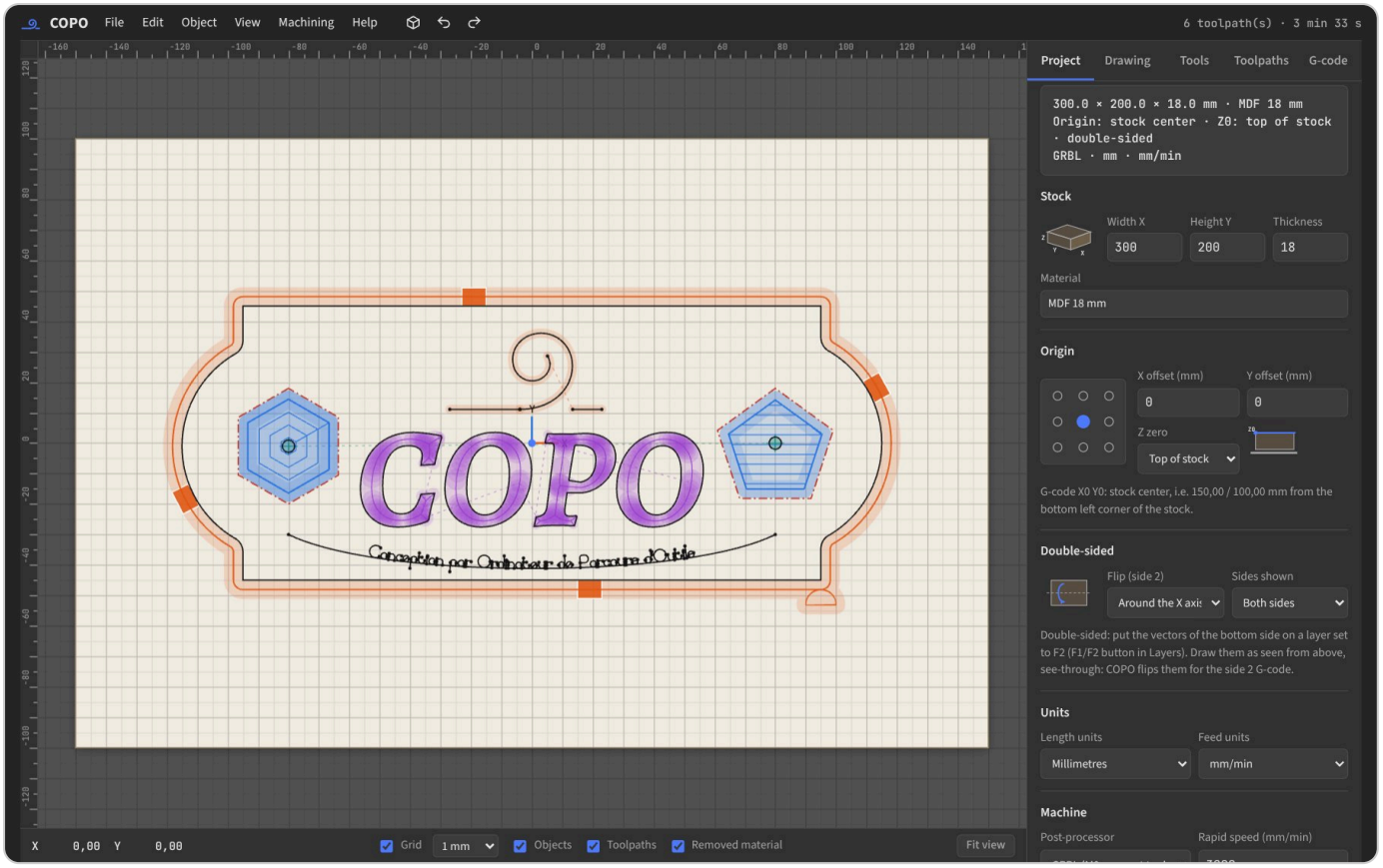


Dogbone: an arc the size of the bit, diagonally in each corner.



T-bone: the arc extends the longer side.

Project tab



The Project tab gathers everything you set at the start of a job. The summary at the top lets you check it all at a glance.

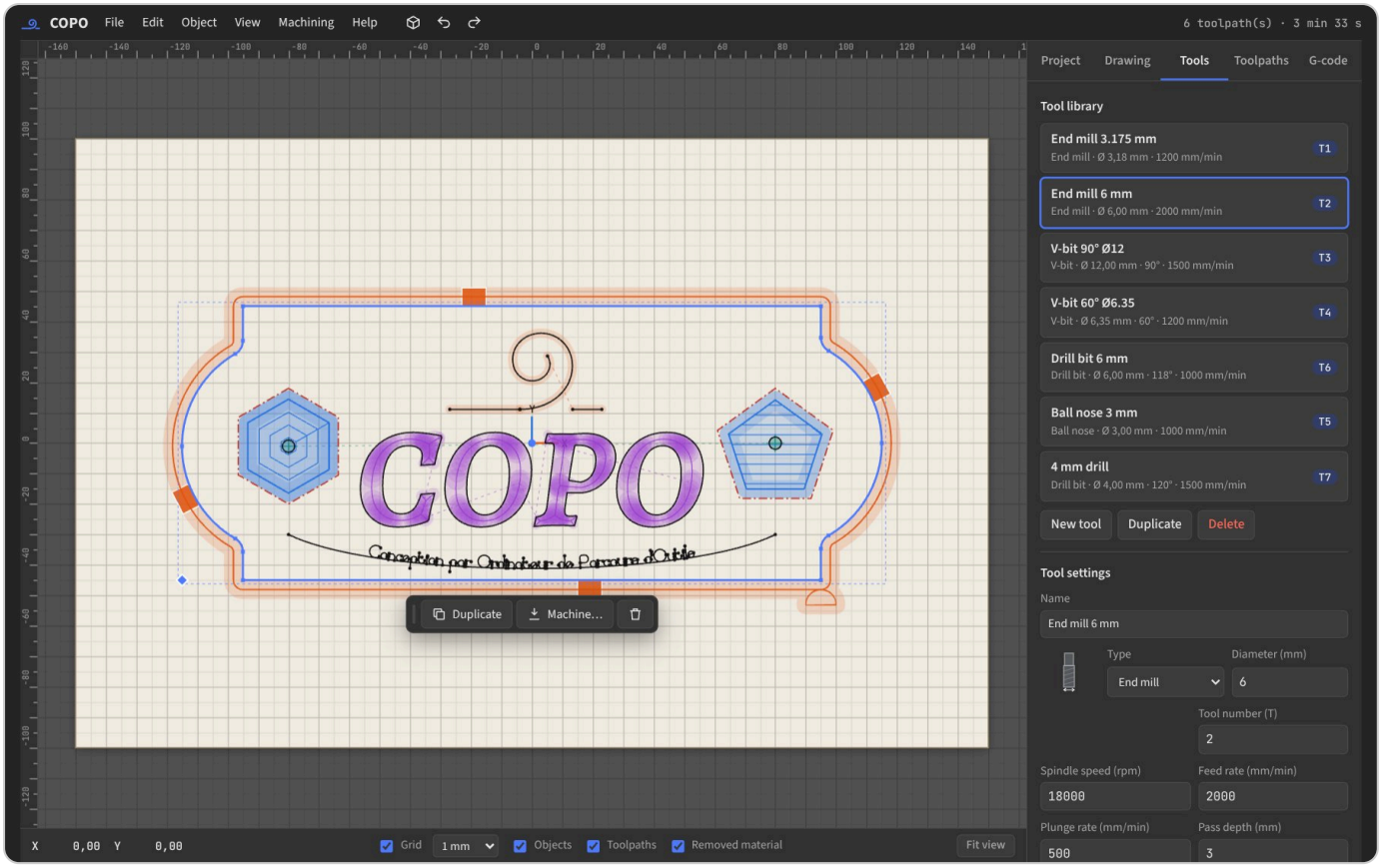
The **Project** tab, first in the right panel, opens by itself when you create a new project. It contains:

- **Stock**: X width, Y height, thickness and material (used by the setup sheet);
- **Origin**: the X0 Y0 point on the nine-point grid, an X/Y offset to place it outside the stock, and Z zero on the top of the stock or on the table;
- **Double-sided**: the flip axis and the sides shown;
- **Units**: millimetres or inches, feed unit;
- **Machine**: post-processor, rapid speed, safe and clearance Z, G2/G3 arcs and comments.

The origin point must match the point where you set X0 Y0 on the machine. Stock, origin and material are saved with the project; units and machine settings are remembered in the browser.

> Example: with the origin at the bottom left corner, a square whose corner sits at X0 Y0 on the canvas is cut from the bottom left corner of your sheet. The line under the grid tells you where the G-code X0 Y0 falls.

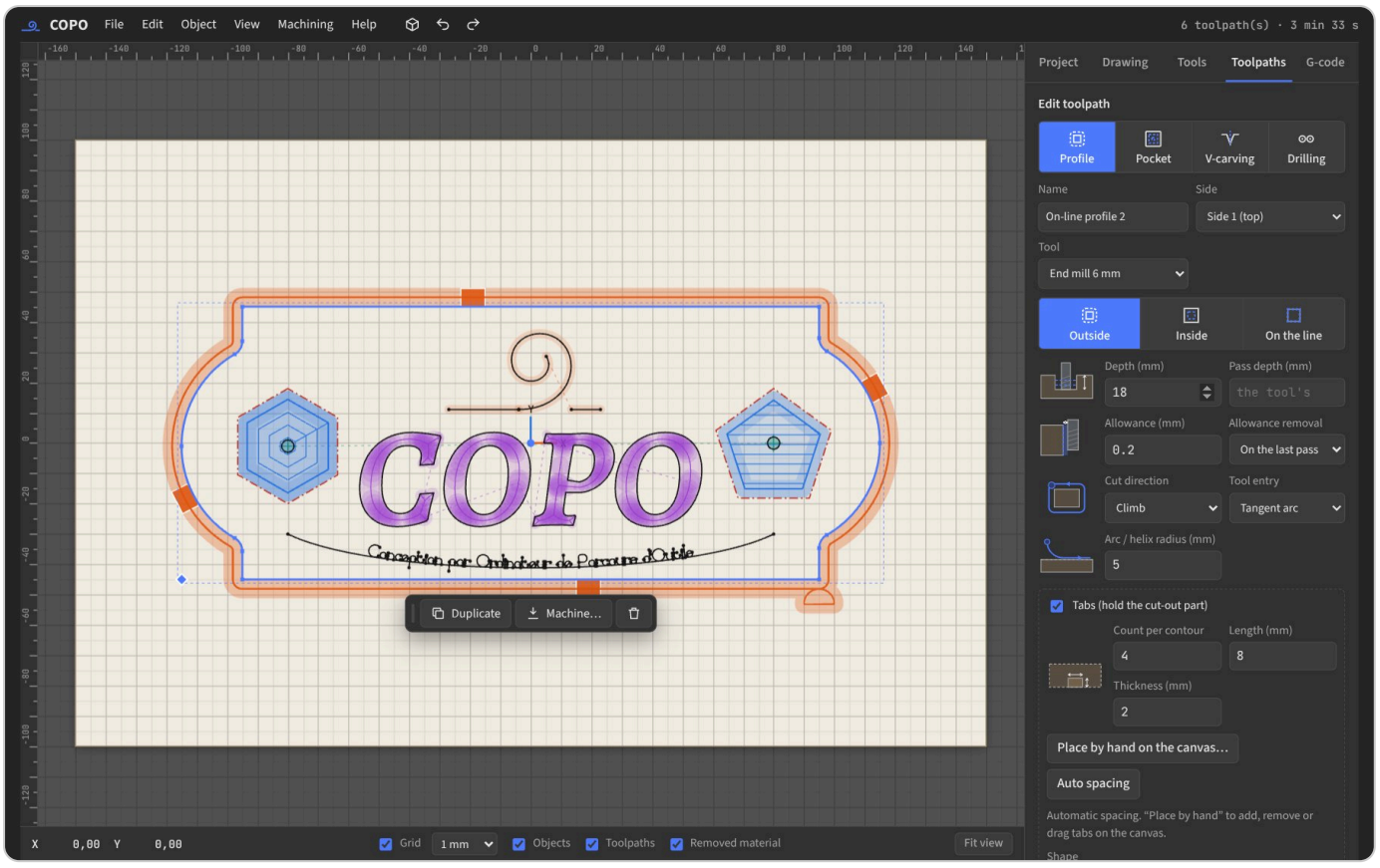
Tool library



The library is kept in the browser and reused in every new project. Export it to a file to back it up or share it.

SETTING	PURPOSE
Type	End mill, ball nose, V-bit (with the V angle) or drill bit (with the point angle, usually 118°).
Diameter	Used to calculate offsets and pockets.
Tool number (T)	Written in the G-code of tool-change post-processors (`T2 M6` , `M6 T2`).
Spindle speed	In rpm, written with `M3` .
Feed, plunge rate	In mm/min, for cutting and for plunging or ramping.
Pass depth	Maximum depth per pass; a toolpath can override it.
Stepover	Percentage of the diameter that overlaps between two pocket passes.

Toolpaths



Click a toolpath in the list to edit it: its path is highlighted on the canvas. The text under the form sums up what the machine will do.

The four types

- **Profile**: outside (the part is kept), inside (the hole is cleared) or on the line. On an open line, the tool can run left or right of the path, in the direction it was drawn.
- **Pocket**: in *rings* parallel to the contour or as a *raster* at the chosen angle. *Rest machining* uses a smaller bit for the corners a larger bit could not reach: keep the same depth and place this toolpath after the original pocket.
- **V-carving**: the V-bit follows the medial axis of the path; it goes deeper where the stroke is wide and rises up to the tip of every corner. A maximum depth gives a flat bottom.
- **Drilling**: one hole at the center of each closed shape, with pecking and a dwell at the bottom. Circles larger than the bit are milled helically to the drawn diameter. With a drill bit, each hole is drilled at the bit diameter; COPO shows the tip length to add for a through hole.

Passes, allowance and cut direction

The total depth is split into passes of the chosen depth. An **allowance** leaves a little material on the walls while roughing; it is removed on the last pass or by a full-depth finishing pass. The cut direction is *climb* or *conventional*.

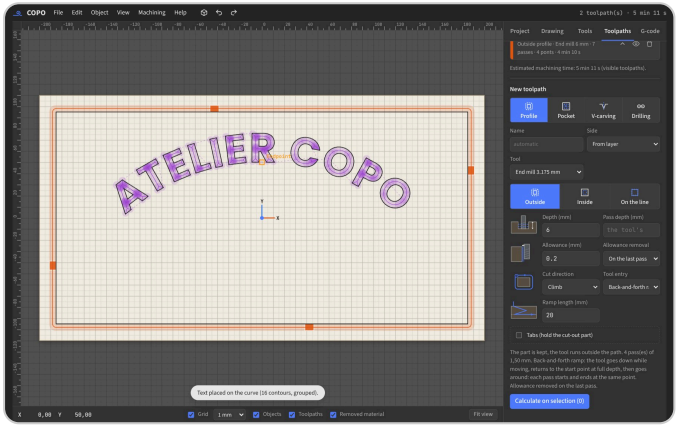
Tool entry

ENTRY	BEHAVIOUR
Back-and-forth ramp	The tool goes down while moving, returns to the start at full depth, then goes around. Each pass starts and ends at the same point. Default setting.
Continuous ramp	The tool goes down while moving and continues around; the ramp portion is recut at the end.
Tangent arc, ramped arc	Arc lead-in and lead-out on the contour, to avoid a mark on the finished wall.
Helix	The tool spirals down, ideal for pockets and holes.
Straight plunge	Vertical plunge, for center-cutting bits or V-carving.

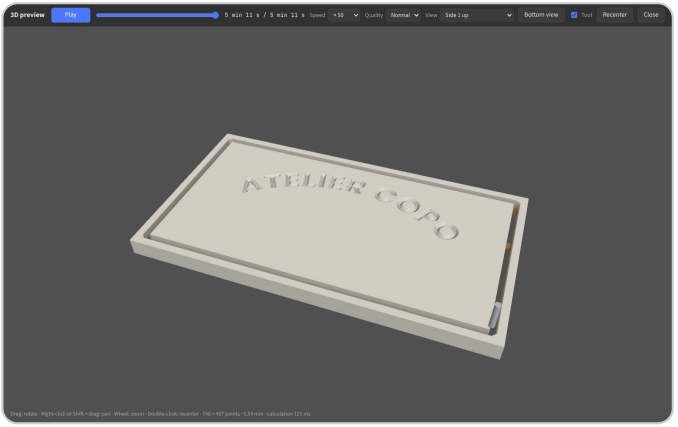
Tabs

Tabs hold a cut-out part until the end. Set their count per contour, length and thickness, flat or triangular (easier to break). **Place by hand on the canvas…** lets you add a tab with a click on the toolpath, drag it or remove it.

Text and V-carving



Text placed on an arc, then V-carved.



The same job in the 3D preview.

Choose a font in the **Text** block list: installed fonts (Chrome and Edge) or your fonts folder (**File** › **Open a fonts folder…**), with a preview. Two boxes at the top of the list, *COPO fonts* and *Computer fonts*, shorten the list; the choice is remembered. Take the **Text** tool `T` and click on the canvas. With **On a curve** placement, click a contour: the text centers on it, above or below, with the spacing you want. The text becomes grouped vectors, ready for V-carving, pocketing or profiling.

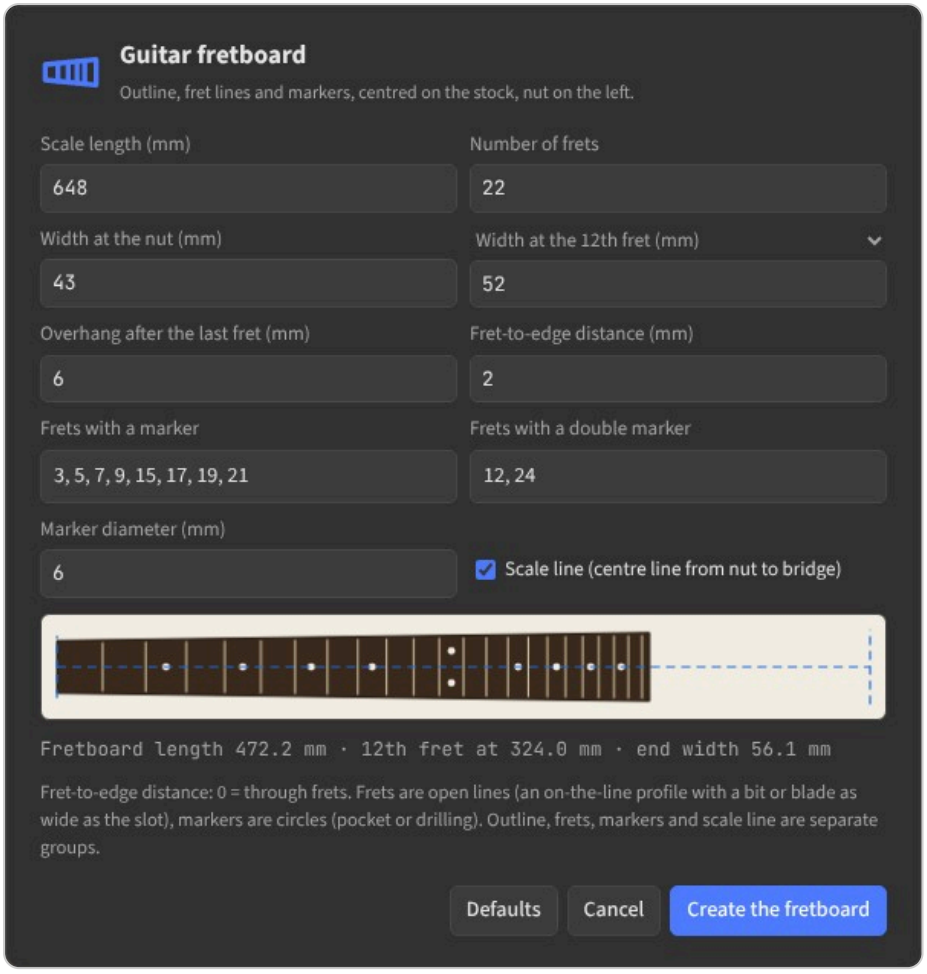
Single-line fonts

COPO comes with about sixty **single-line fonts** (Hershey, EMS, Cutlings, Relief SingleLine, Routed Gothic…), stored in the `fonts/trait-unique` folder and listed first in the font list, with a preview. Each letter is a single stroke: the text is engraved in one pass with a **Profile On the line** toolpath, with a V-bit or a small end mill. It is the fastest way to mark part numbers, plates and assembly marks. When the selection holds only open strokes, **Machine…** offers this profile directly. The `fonts/apercu-polices.html` catalogue (link “See the single-line font catalogue” in the Text block) shows them all with a preview of your text. You can also add your own SVG fonts with **Add fonts…**.

Lutherie: fretboard and inlay

Guitar fretboard

Object › **Guitar fretboard…** draws the fretboard, nut on the left, centred on the stock. Set the **scale length**, the **number of frets**, the **width at the nut** and the second width, *at the 12th fret* or *at the end of the scale*, the overhang after the last fret, the **frets with a marker** and **a double marker** (lists of numbers) and the marker diameter. The **scale line** (ticked by default) draws the centre line from the nut to the bridge, with a mark at each end, to place the bridge. The **fret-to-edge distance** stops the fret lines short of the board edge (0 = through frets). The preview updates live.



The **Guitar fretboard** window with its preview: board, frets, markers and the dashed scale line.

Fret positions

COPO uses the exact equal-temperament formula, the one used by reference fret calculators:

> $\mathbf{d_n = L - L / 2^{n/12}}$ i.e. $d_n = L \times (1 - 2^{-n/12})$

where L is the scale length, n the fret number and d_n the distance from the nut to the fret. Each fret shortens the string by a semitone, a factor of $2^{1/12} \approx 1.05946$; the 12th fret falls exactly at half the scale. The “rule of 17.817” gives the same result; the old, less accurate rule of 18 is not used. No compensation is applied to the frets: intonation is set at the bridge, moved back a few millimetres from the scale line.

FRET	1	5	7	12	
Distance from the nut (648 mm scale)	36.37 mm	163.51 mm	215.64 mm	324.00 mm	466.1(

The result is separate groups: the scale line (not to be machined), the outline, the fret lines (open lines, to cut with an **On the line** profile and a bit or blade as wide as the slot) and the markers (circles, pocketed or drilled).

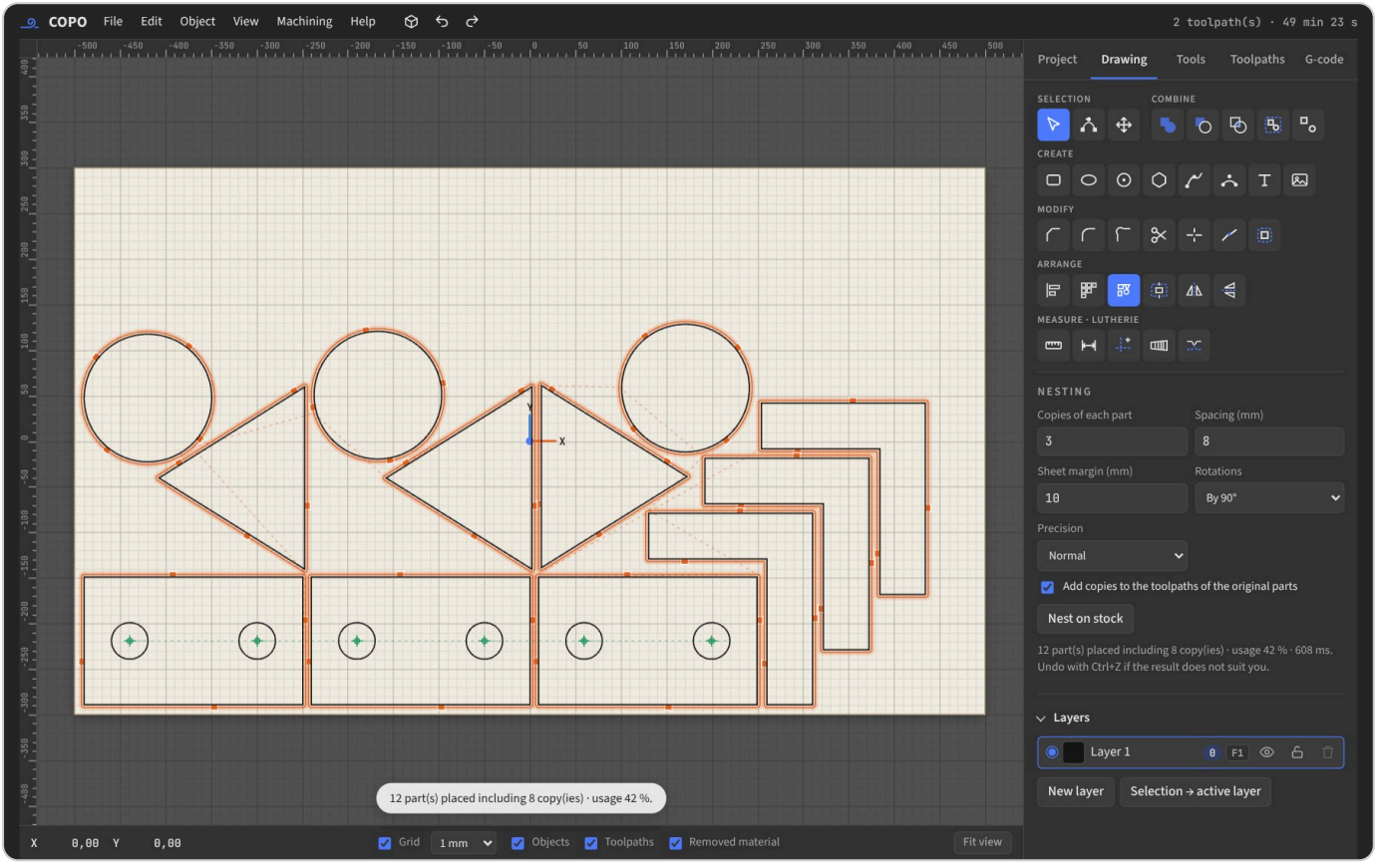
V-carve inlay

Select the design (closed shapes), then **Object › V-carve inlay…**. Choose the V-bit, the **female pocket depth**, the **glue gap** and the **male cut depth**. COPO creates:

- a *female* V-carving of the design in your part;
- the *male* plug: the mirrored design inside a frame, on the “Inlay · male” layer, with a V-carving that starts at the pocket depth minus the gap. Its toolpath is created *hidden*: to cut it from the inlay wood, show it and hide the female one before saving the G-code.

Flip the plug, glue it into the female pocket, then plane or sand off the excess. The V-carving **Start depth** setting is also available by hand in the toolpath form.

Nesting



Four parts in three copies on a 1000 × 600 mm sheet. The L shapes interlock and the copies join the toolpaths of the original parts.

Object › **Nesting…** opens the block in the Drawing tab. Select the parts (with no selection, the whole drawing is used) and set the number of copies, the spacing, the sheet margin, the allowed rotations and the precision. COPO places the parts following their true shape, largest first, as far down and left as possible. A group or a text stays in one block. Parts that do not fit stay selected, and the sheet usage is shown.

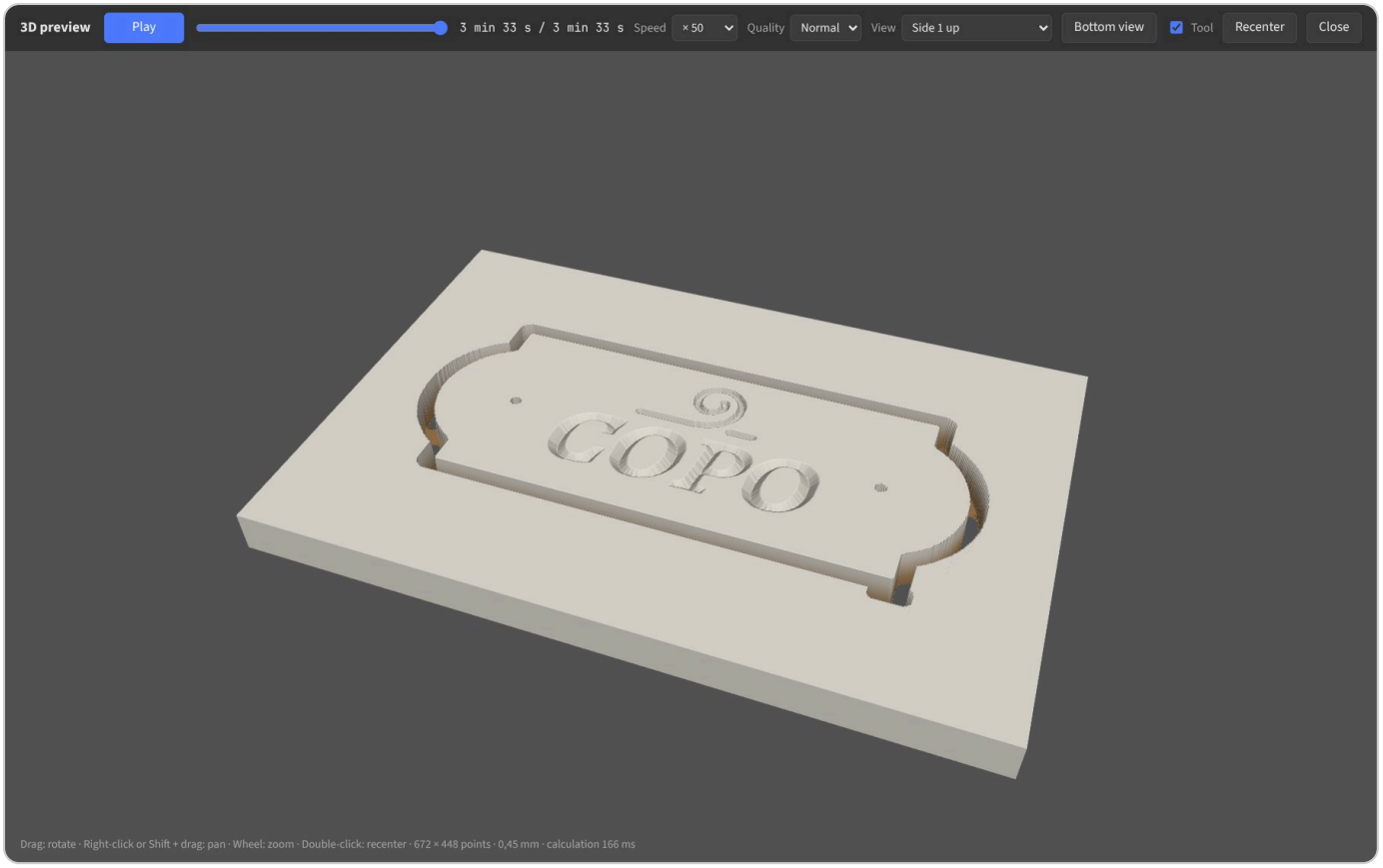
> For cut-outs, use a spacing at least equal to the bit diameter. `Ctrl` + `Z` undoes the nesting if you don't like the result.

Double-sided machining

1. Put the vectors of the bottom side on a layer set to **F2** (F1/F2 button in the layer list). Draw them as seen from above, see-through.
2. Choose the flip: around the X axis (top ↔ bottom) or the Y axis (left ↔ right).
3. Side 2 toolpaths are calculated on the flipped geometry; the cut direction is kept.
4. In the G-code, side 1 is followed by an `M0` pause: turn the part over, set X0 Y0 Z0 again at the same point and resume.

The 3D preview shows both sides; the **Side 2 up** option shows the part flipped.

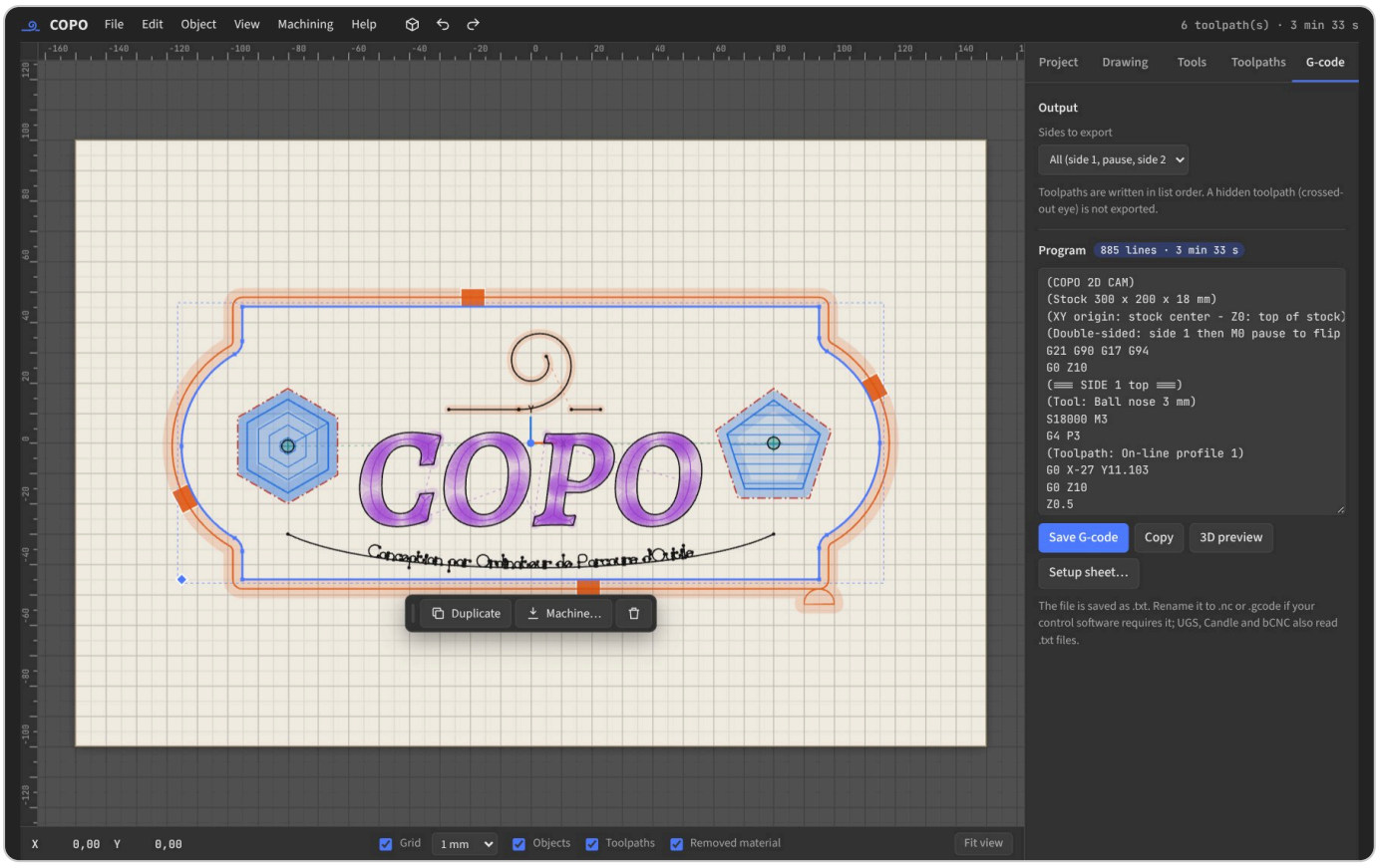
3D preview



Drag to rotate, right-click or Shift to pan, wheel to zoom, double-click to recenter.

The preview simulates material removal through the real stock thickness, through-holes included. The **Play** button replays the job at the chosen speed and the slider jumps to any moment. The preview needs WebGL 2 (a recent Chrome, Edge or Firefox).

G-code



Toolpaths are written in list order; a hidden toolpath is not exported.

POST-PROCESSOR	TOOL CHANGE
GRBL	`M0` pause between two tools.
GRBL manual	Before each tool, the first one included: retract with `G53 G0 Z-10` (adjustable) and `M0` pause. Also a `G53` retract at the end of the program.
Carbide Motion, Onefinity	`M6 T…` : the control software asks you to change the tool.
LinuxCNC, Mach3/4, UCCNC, MASSO	`T… M6` with the tool number, `G40 G49 G80` header, arcs with `G91.1` . `.ngc` file (LinuxCNC), `.tap` (Mach, UCCNC) or `.nc` .
Custom	You write the header, tool change and program end, with variables: `{UNITS}` , `{SAFEZ}` , `{TCZ}` , `{T}` , `{S}` , `{D}` , `{TOOL}` . File extension and `G91.1` are up to you.

```
(Load tool: End mill 6 mm D6 - set Z zero again then resume)
G21 G90 G17 G94
G53 G0 Z-10
M0
S18000 M3
G4 P3
G0 X10 Y10
G0 Z10
…
M5
G53 G0 Z-10
G0 X0 Y0
M30
```

The post-processor, rapid speed (for the time estimate), safe Z, clearance Z, comments and G2/G3 arcs are set in the Project tab, Machine block. The file gets the extension the post-processor expects (`.nc` , `.tap` , `.ngc`).

> Always check the G-code with a dry run, tool above the stock, before cutting.

Units: millimetres or inches

In the Project tab, Units block, choose the **Length units** (millimetres or inches) and the **Feed units** (mm/min, mm/s or in/min). The choice applies everywhere: drawing, rulers, grid (1/16, 1/8, 1/4 and 1 inch), coordinates, dimensions, tools, toolpaths and setup sheet. In inches, one option shows values as fractions (to 1/64) and another writes the G-code in inches (`G20`) for controllers set up that way.

The project is always saved in millimetres: you can switch units at any time without distorting anything, and a project opens correctly for someone working in the other unit. The choice is remembered in the browser.

Setup sheet

The **Setup sheet…** button in the G-code tab (also in the Machining menu) prepares a two-page setup sheet for the workshop. Enter the part name, the material and your notes; the sheet contains:

- the stock, X0 Y0 origin, Z zero, flip and post-processor;
- a summary: estimated time, maximum depth, machined extent from the origin, cutting length and removed material volume;

- the 2D view of each side with toolpaths, origin and stock dimensions, and two dimetric views of the simulation (side 1 and side 2);
- the tool loading order, then the toolpath table: operation, tool, depth, passes, allowance, entry, tabs, feed, plunge and time;
- the points to watch it detects (cut through the stock, cut-out without tabs, pass too deep, job outside the stock, hidden toolpaths);
- a checklist before starting, a notes area and sign-off boxes.

Print it or save it as a PDF, or keep the HTML file with the G-code. The sheet settings are saved with the project.

Install and use offline

- **Online**: open the GitHub Pages site of the COPO repository. Nothing to install.
- **Offline**: download the complete COPO folder (“Code › Download ZIP” on GitHub) and unzip it. Open `index.html` with a double-click: everything is already in the `lib` folder (calculation libraries and interface fonts) and in the `fonts` folder. Nothing else to install or download.
- **Interface fonts**: the `lib/fonts` folder holds the COPO fonts (Source Sans 3, JetBrains Mono, Barlow Condensed). Offline, the interface, the setup sheet and this guide keep the same look.
- **Engraving fonts**: put your `.ttf`, `.otf` or `.woff` files in a `fonts` folder next to COPO.html and open it once from the File menu.

COPO runs on Windows, macOS and Linux, in a recent Chrome, Edge or Firefox. The list of installed fonts is only available in Chrome and Edge.

Keyboard shortcuts

KEY	ACTION	KEY	ACTION
`V`	Select	`Ctrl` + `Z` / `Y`	Undo / Redo
`N`	Nodes	`Ctrl` + `S` / `O`	Save / Open a project
`H` or `Space`	Pan the view	`Ctrl` + `D`	Duplicate
`R` `E` `C`	Rectangle, ellipse, circle	`Ctrl` + `A`	Select all
`O`	Polygon and star	`Ctrl` + `G`	Group (`Shift` : ungroup)
`P` `A` `T`	Pen, arc, text	`Ctrl` + `]` / `[`	Forward / backward (`Shift` : front / back)
`K` `F`	Chamfer, fillet	`Del`	Delete the selection or node
`X` `U`	Trim, cut	Arrows	Move by 1 mm (`Shift` : 10 mm)
`M` `D`	Measure, dimension	`Esc`	Cancel the current path, deselect

On a Mac, `Cmd` replaces `Ctrl`.

Report a bug

Help › Report a bug… menu: describe the problem, then click **Open the report on GitHub**. The report arrives pre-filled with technical information (browser, units, stock, toolpaths, recent errors). The workspace screenshot is copied to the clipboard: paste it into the report with `Ctrl` + `V`. If you tick “Attach the project”, drag the downloaded file into the report. A free GitHub account is enough. Ideas for improvements are welcome too at github.com/gbulleryahen-hash/copo/issues.

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Libraries used, bundled in the `lib` folder: Clipper (Boost Software License), opentype.js (MIT), d3-delaunay and delaunator (ISC).

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